Do Diplomas Deliver?

Bureaucrat Education and Public Service Outcomes

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The Paper

- Indian Civil Service follows Macaulayan legacy → assigns bureaucrats to districts without accounting for programmatic fit.
- Generalist ideals → An engineer or social scientist bureaucrat occupy identical roles, albeit face systematically different tasks depending on which programs dominate the district.
- Educational training creates task-specific comparative advantage within a generalist bureaucracy, and whether the mismatch between officer type and program type has measurable costs.

Research question: Does the educational background of the District Collector affect which public goods get delivered, and how well?

Policy margin: Unlike selection quality or incentive design, posting rules are a zero-cost margin. Reweighting assignments by educational background costs nothing.

Related Literature

- Public goods delivery, selection quality and incentive design. *(Finan, Olken & Pande 2017; Dal Bó, Finan & Rossi 2013; Rasul & Rogger 2018)*.
- Assignment literature *(Roy 1951; Heckman & Sedlacek 1985)*
- Noteworthy : *Xu, Bertrand & Burgess (2018)*, who use the IAS cadre allocation rule to study home-state posting effects.
- **Priority mechanisms & matching theory:**
  Abdulkadiroğlu & Sönmez (2003) — school choice as priority queue;
  Roth & Sotomayor (1990) — stability and efficiency of matching mechanisms
- **Misallocation of resources:**
  Hsieh & Klenow (2009) — misallocation across heterogeneous producers generates aggregate output losses; analogous logic applied to bureaucratic human capital
- **Optimal bureaucrat allocation:**
  Rasul & Rogger (2018) — task clarity and bureaucrat performance;
  this paper: first to formalize IAS queue as priority mechanism and test task-specific comparative advantage

The IAS and the District Collector

Indian Administrative Service:

- Recruited nationally via CSE (UPSC) competitive examination
- Allocated to state cadres for life (caste-based quotas + home preference)
- Generalist by design — officers rotate across portfolios every 1–3 years
- Approximately 5,000 active officers; roughly 640 districts across India

District Collector / District Magistrate:

- Principal administrative officer in a district; responsible for all program implementation
- Oversees PMGSY (rural roads), NREGA (welfare employment), electrification, land administration
- Posting determined by state government: seniority, vacancy, and political context
- No formal rule links educational background to district assignment
- Mean tenure $\approx$ 15–18 months (Iyer & Mani 2012; consistent with our data)

Data

- **IAS administrative records** (Trivedi Center): DC posting spells for 4,348 officers, 1985–2008; designation, office, start/end dates, educational degree, cadre, allotment year
- **DMSP nightlights** (NOAA): Annual satellite-derived calibrated light output, all 640 Census 2011 districts, 1994–2008; fully observed, no missing values
- **PMGSY roads** (Ministry of Rural Development): Annual district rural road completions (km); coverage expands from 36 districts in 2001 to $\sim$340 by 2002–2008; restricted to 2001–2008
- **Economic Census employment** (MoSPI): Annual non-farm establishment employment, 553–640 districts; to isolate post-NREGA period
- **Census 2011** (RGI): SC/ST population share, literacy rate, total population; time-invariant controls absorbed by district fixed effects

Educational Composition of DC Officers

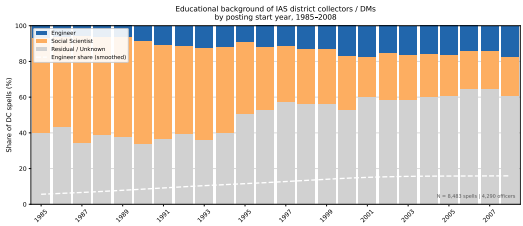
Group	Spells	Share
Social Scientist	873	36.8%
Engineer	486	12.2%
Residual	528	22.2%
Unclassified	870	28.8%
Total (classified)	1,887	

Engineer: Engineering, Technology & Mathematics.

Social Scientist: Social Science, Liberal Arts, Governance & Law.

Residual: Natural Sciences, Medical.

Unclassified: N.A. or missing education record.



Engineer share remains stable at roughly 7–12% throughout the sample period. Social scientists account for 45–55% of classified spells. Variation across time and states provides identifying variation in officer education type.

Comparative Advantage in Program Delivery

Setup: Officer i of education type $\tau \in \{T, S, R\}$ produces output in program $k \in \{\text{infrastructure}, \text{welfare}\}$:

$$Y_{idt} = \eta(\tau_i,\ k_d) + \alpha_d + \lambda_{st} + \varepsilon_{idt}$$

where $\eta(\tau, k)$ is the implementation efficiency of type τ on program k.

Predictions:

P1 $\eta(T, \text{infra}) > \eta(S, \text{infra})$: engineer advantage on infrastructure programs

P2 $\eta(S, \text{welfare}) > \eta(T, \text{welfare})$: social scientist advantage on welfare programs

P3 Engineer advantage increasing in district infrastructure program intensity

Estimating Equation

$$Y_{dt} = \beta_1 \cdot \text{Engineer}_{it} + \beta_2 \cdot \text{Residual}_{it} + \gamma X_{dt} + \alpha_d + \lambda_{st} + \varepsilon_{dt}$$

- Y_{dt}: outcome in district d, year t
- Social Scientist is the omitted baseline throughout
- β_1: engineer effect relative to social scientist
- α_d: district fixed effects; λ_{st}: state $\times$ year fixed effects
- X_{dt}: officer tenure, seniority at posting, election year indicator
- Standard errors clustered at district

P1 predicts $\beta_1 > 0$ on infrastructure outcomes. P2 predicts $\beta_1 < 0$ on welfare outcomes.

Identification via the Seniority Queue

Assignment Rule

$$i^*(d,t) = \arg\max_{i \in \mathcal{A}_{st}} r_i, \qquad \mathcal{A}_{st} = \{i \in \mathcal{I}_s : a_{it} = 1\}$$

- Priority r_i fixed at IAS entry — independent of district characteristics
- Education type e_i does not appear in the assignment rule
- Availability a_{it} depends on state-wide rotation history, not district d

Key Independence Result

$$e_{i^*(d,t)} \perp \mathbf{X}_d \,\Big|\, r_{i^*(d,t)},\, \mathcal{A}_{st}$$

- Education of assigned officer independent of district characteristics
- Verified empirically: joint $F = 0.89$, $p = 0.470$ (Table 1, Panel B)

Matched Analysis Sample

Merge: IAS DC spells $\xrightarrow{\text{fuzzy name match}}$ district crosswalk $\xrightarrow{\text{pc11 ID}}$ panel outcomes

Total DC spell-years, 1985–2008	8,494
Matched to Census 2011 district ID	5,612 (66.1%)
Within panel window, 1994–2013	5,670
Matched to outcome row	3,836 (67.7% of in-panel spells)
Analysis sample (classified officers only)	
Spell-years	1,887
Unique DC spells	3,355
Districts	422 of 640 (66%)
Years	1994–2008

Unmatched: blank office field (553 spells), department-level postings, or districts absent from crosswalk. Unclassified education (N.A.) excluded from regression sample; robustness to inclusion available on request.

Table 1: Balance

Panel A: Officer characteristics

Group	N spells	Tenure (mo.)	Allot. year	% Female
Engineer	578	17.7	1992.6	4.2
Social Scientist	1,114	15.8	1990.5	13.6
Residual	698	17.1	1990.8	10.2

Panel B: District characteristics (pre-treatment Census values)

Characteristic	Engineer	Soc. Sci.	Difference	p-value
SC share	0.156	0.159	-0.002	0.534
ST share	0.137	0.120	$+0.018^{*}$	0.084
Literacy rate	0.525	0.532	-0.007	0.260
Population ('000)	1,855	1,940	-85	0.355
Joint F-test			0.89	0.470

Panel A: spell-year means. Panel B: district-level means from Census 2011. Difference = Engineer − Social Scientist. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 3: Infrastructure Outcomes

	Panel A: Nightlights per capita, 1994–2008			Panel B: Road completion rate, 2001–2008		
	(1)	(2)	(3)	(4)	(5)	(6)
Engineer	-0.0007	-0.0007	-0.0010	0.0284	0.0233	0.0158
	(0.0008)	(0.0008)	(0.0008)	(0.0402)	(0.0411)	(0.0245)
Residual	0.0002	0.0004	-0.0000	0.0007	0.0196	0.0412
	(0.0003)	(0.0004)	(0.0005)	(0.0379)	(0.0387)	(0.0268)
District FE	✓	✓	✓	✓	✓	✓
Year FE	✓	✓		✓	✓	
State×Year FE			✓			✓
Controls		✓	✓		✓	✓
N	2,390	2,274	2,274	596	586	586
R^2	0.001	0.002	0.406	0.004	0.034	0.498
Control mean	0.019	0.019	0.019	0.369	0.369	0.369

Baseline: Social Scientist DCs. Panel A: DMSP light / population. Panel B: roads completed / total road length (km), 2001–2008. Controls: tenure, allotment year, incumbent win. SE clustered at district. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Engineer effect on roads positive across all specs; switches on with PMGSY (2001). Nightlights near zero — consistent with comparative advantage averaging across program types.

Table 4: Employment per Capita

	Employment per capita (ec_emp_all / tot_p)		
	(1)	(2)	(3)
Engineer	0.100***	-0.000	-0.000**
	(0.015)	(0.000)	(0.000)
Residual	0.124***	-0.000	-0.000
	(0.017)	(0.000)	(0.000)
District FE	✓	✓	✓
Year FE	✓	✓	
State×Year FE			✓
Controls		✓	✓
N	438	433	433
Within R^2	—	0.000	0.000
Control mean	0.076	0.076	0.076

Sample: 2006–2008 (post-NREGA). Baseline: Social Scientist. Controls: tenure, allotment year, incumbent win. SE clustered at district. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Engineer coefficient negative in preferred spec (Spec 3, $p = 0.031$). Sign reversal rules out generic ability story.

Summary of Estimates

Engineer coefficient ($\hat{\beta}_1$) relative to Social Scientist baseline, preferred specification (Spec 3):

Outcome	$\hat{\beta}_1$	p-value
Road completion rate	$+0.022$	0.364
Nightlights per capita	≈ 0.000	0.921
Non-farm employment p.c.	-0.000^{**}	0.031

Engineer DCs show positive coefficients on infrastructure and negative on welfare-adjacent outcomes relative to Social Scientist DCs — directionally consistent with P1 and P2 across all specifications.

A generic ability story predicts $\hat{\beta}_1 > 0$ uniformly. Sign reversal across program types is inconsistent with that hypothesis.

Nightlights null consistent with comparative advantage: aggregate satellite luminosity averages infrastructure and welfare effects, which attenuate toward zero under P1 and P2.

Instrument Validation

	Road completion rate		Nightlights per capita	
	OLS	2SLS	OLS	2SLS
	(1)	(2)	(3)	(4)
Engineer	0.0281	0.0265	-0.0008	0.0386
	(0.0249)	(0.1993)	(0.0007)	(0.0390)
First-stage F	—	6.94 ×	—	6.18 ×
Observations	453	453	2,339	2,339
District FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓

Notes: Baseline category: SocialScientist. Instrument (2SLS): Queue_refined = opposition_turnover × cadre_queue_pct, where opposition_turnover = 1 when incoming state government is from a different party than outgoing, and cadre_queue_pct $\in [0, 1]$ is officer's percentile position in the cadre's seniority queue in that year (0 = most senior). Cols (1)–(2): roads subsample, year $\geq$ 2001. Cols (3)–(4): full panel, 1994–2008. Clustered SEs at district level. ✓: $F > 10$; ×: $F \leq 10$. Stars: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

- **First stage:** Queue_refined $\rightarrow$ Engineer, $F = 6.94$ (roads), $F = 6.18$ (full panel)
- **IV/OLS ratio = 0.94** for roads: strategic assignment bias is empirically small
- **Why $F < 10$:** opposition-turnover years are sparse (20 % of observations) — longer panel restores power

Robustness

	R1: tenure $\geq$ 6 mo.	R2: 1994–2004	R3: state cluster	R4: pure-ID districts
	(1)	(2)	(3)	(4)
Panel A: Nightlights per capita				
Engineer	0.000	-0.000	-0.001	-0.001
	(0.000)	(0.000)	(0.001)	(0.001)
N	1,933	1,687	2,274	2,068
Panel B: Road completion rate				
Engineer	0.031	0.037	0.023	0.023
	(0.058)	(0.160)	(0.038)	(0.040)
N	481	244	586	515

All specs: district FE + year FE + controls. Baseline: Social Scientist. R1: tenure $\geq$6 mo. R2: 1994–2004. R3: state clustering. R4: pure-ID districts. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Sign and magnitude stable across: short-tenure exclusion, pre-2005 sample, state-level clustering, pure-ID-district restriction.

Next Steps

- **Heterogeneity (P3) — testable now**
  Engineer × high-PMGSY-intensity districts: does the infrastructure effect concentrate where program pipeline is largest?

- **NREGA MIS outcomes.** Acquire program-specific person-days by district-year. Economic Census non-farm employment is an imperfect proxy for P2; NREGA MIS is the correct test.

- **Extend panel.** DMSP available through 2013; VIIRS from 2012. Extension adds five years and substantially improves power, particularly for the roads and NREGA samples.

- **Optimal allocation extension**
  Estimated complementarities motivate a mechanism design question: what two-stage assignment rule — seniority determines eligibility, education determines assignment within tier — achieves first-best under our parameters? (Abdulkadiroğlu & Sönmez 2003; Roth & Sotomayor 1990)

The Optimal Matching Problem

Current Mechanism: Seniority Queue

$$i^*(d,t) = \arg\max_{i \in \mathcal{A}_{st}} r_i \qquad \text{[ignores match quality]}$$

Optimal Two-Stage Mechanism

$$i^*(d,t) = \arg\max_{\substack{i \in \mathcal{A}_{st} \\ r_i \geq \bar{r}}} \eta(e_i, \tau_d) \qquad \text{[seniority then match quality]}$$

- Stage 1: seniority threshold $\bar{r}$ preserves career incentives
- Stage 2: within eligible pool, assign engineer $\rightarrow$ infra district, social scientist $\rightarrow$ welfare district
- Stable in the sense of Roth & Sotomayor (1990): no officer-district pair mutually prefers to deviate

Conclusion

Question: Does educational background of the District Collector affect public goods delivery across program types?

Preliminary findings:

- Engineer and social scientist DCs posted to observationally equivalent districts: joint $F = 0.89$, $p = 0.470$
- Engineer DCs show positive coefficients on road completion rates relative to social scientist baseline across all specifications
- Engineer DCs show negative coefficients on non-farm employment per capita, significant at 5% in preferred specification ($p = 0.031$)
- Pattern is directionally consistent with comparative advantage; inconsistent with a generic ability explanation

Policy implication: A posting rule incorporating educational background could improve program delivery at zero additional fiscal cost.